

Miguel M. G. Pascual-Caballo

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Summary

PhD student in Fluid Dynamics at ICMAT. My research focuses on the analysis of existence and instability of solutions to partial differential equations in fluid dynamics, using analytical and computer-assisted techniques.

Education

- 2022 – present **Instituto de Ciencias Matemáticas (ICMAT)** – Madrid, Spain
PhD in Mathematics (Fluid dynamics), Universidad Autónoma de Madrid (UAM)
Advisor: Ángel Castro
- 2021 – 2022 **Universidad Complutense de Madrid** – Madrid, Spain
Master in Mathematics (Matemáticas Avanzadas) Master's thesis: De Rham theorems in co-homology of manifolds
- 2017 – 2021 **Universidad Complutense de Madrid** – Madrid, Spain
Bachelor in Mathematics Bachelor's thesis: Geometric mechanics of the rigid body

Research visits

- 2025 **Brown University** – Providence, RI, USA
Visiting researcher (3 months), hosted by Javier Gómez-Serrano

Publications

- 2026 M. M. G. Pascual-Caballo. *On the instability of De Gregorio excited states.* Preprint, [arXiv:2610.03287](https://arxiv.org/abs/2610.03287)
- 2026 Á. Castro, J. Gómez-Serrano, and M. M. G. Pascual-Caballo. *Linear instability of a Burgers–Hilbert traveling wave.* Preprint, [arXiv:2605.03920](https://arxiv.org/abs/2605.03920)
- 2025 M. M. G. Pascual-Caballo. *Stationary homogeneous solutions for the inviscid SQG equation.* Preprint, [arXiv:2510.03108](https://arxiv.org/abs/2510.03108)

Talks

- 2026 **Linear instability of a Burgers–Hilbert traveling wave**
Partial Differential Equations and Applications Seminar, IME-USP, São Paulo, Brazil
- 2025 **A Brief Introduction to Computer-Assisted Techniques in Analysis**
BYMAT, Sevilla, Spain
- 2024 **Linearization and Stability in Fluid Dynamics**
BYMAT, Valladolid, Spain
- 2023 **Compactness: Geometry and Topology**
Junior Colloquium (PhD student seminar), Universidad Autónoma de Madrid

Teaching

- 2024 – 2025 Teaching assistant, Mathematics I (Biomedical Engineering)
Universidad Autónoma de Madrid
- 2024 – 2025 Teaching assistant, Algebra (Computer Engineering)
Universidad Autónoma de Madrid
- 2023 – 2024 Teaching assistant, Calculus II
Universidad Autónoma de Madrid

Technical skills

Programming

Python (SageMath)

Mathematical software

MATLAB, Mathematica

Languages

English (Upper Intermediate)

Honors and scholarships

- 2022 JAE Summer School, ICMAT
- 2017 Bronze Medal, Spanish Mathematical Olympiad